

FAI Governance for Observational Studies

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

The Full Aspect Integration (FAI) architecture governs inter-Self coordination events in which participating CKS-governed Selves exchange substrate content under joint human authority. A specific and distinct variant arises when an FAI event is structured as an observational study: participants consent to contributing their governance patterns for collective governance research, with the primary objective being research data generation rather than operational coordination outcomes alone. This note formalizes the governance architecture for observational study FAI events. It covers five components: (1) research purpose stated in the FAI configuration itself; (2) participant consent as governed substrate content in each Self's home governance — a specific form of propagation permission, not a verbal agreement; (3) the research record as an additional governance artifact beyond the standard seventeen governance record categories; (4) anonymization governance requirements in the cross-organizational agreement; and (5) the connection between accumulated observational study FAI events and the empirical validation of Claim 6's population-scale collective evolution projection, acknowledged as downstream future work. The note closes with the anti-pattern — ungoverned research use — and an operational test for whether a given observational study FAI event satisfies the governance requirements.

1. Why this derivation is needed

Paper 3 (Li, April 2026) establishes that accumulated FAI events across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope (Claim 6, §9). The architectural commitment is stated confidently; the paper is explicit that specific population-scale dynamics await empirical observation, and that empirical validation at scale is downstream work. This is the calibrated-humility register Paper 3 applies at Claim 6's scope:

the design pattern is defended, and the empirical question of what that pattern produces at population scale is left open.

The question left open has an architectural answer that the paper does not develop: what governed mechanism would produce the empirical data needed to answer it? Observational study FAI events are that mechanism. They are the architectural path through which population-scale governance dynamics could eventually be studied under the same accountability properties the architecture provides for operational coordination. D2.68 formalizes the governance architecture for this specific FAI variant.

A second motivation is that FAI governance records — the seventeen categories established in D2.18 — belong to the participating Selves. They are the Selves' substrate content, produced in the course of governed coordination. Repurposing those records for research without explicit authorization from the governing humans of each participating Self would violate the architecture's foundational property: that substrate content remains under the authority of those who govern it. Formalizing the research-use authorization as a distinct governance requirement closes this gap.

2. The observational study FAI configuration

Every FAI event is fully configured before execution — the FAI configuration specifies what participating Selves contribute, under what terms, and for what purposes. This configuration is itself substrate content under joint authority across participating Selves' governance (D2.12; Paper 3 §8). In a standard coordination or knowledge transfer event, the purpose specified in the configuration is operational: the event is designed to produce coordination outcomes — shared aspects, resolved conflicts, updated governance records.

An observational study FAI event differs from a standard event at the level of purpose. The FAI configuration explicitly specifies that a primary objective is research data generation about governance patterns. This is not a secondary annotation added after the fact; it is a configuration parameter, authored as substrate content under joint authority before the event begins. Because the configuration is itself governed, the research purpose is governed: the participating Selves' authority structures have jointly authorized that the event will serve research purposes in addition to any operational coordination purposes.

This configuration-level specification matters for two reasons. First, it makes the research purpose inspectable and auditable — an observer reviewing the shared substrate governance artifacts can locate the research-purpose statement as configuration content, with full provenance. Second, it anchors subsequent consent requirements to a governed artifact: each participating Self's research-use authorization (§3 below) is authorization of a purpose that is stated as substrate content, not authorization of an informally described activity.

Observational study FAI events can be structured across the full range of FAI cardinalities — two-Self events studying dyadic coordination patterns, multi-Self events studying network-level dynamics, and sequences of FAI events studying longitudinal evolution. The research purpose configuration applies at each event regardless of cardinality.

3. Participant consent as governance authorization

The most consequential governance requirement in an observational study FAI event is also the most easily misread: each participating Self must provide explicit research-use authorization before its contributed governance patterns can be used for research purposes.

This authorization is not a verbal agreement, an informal consent form, or a procedural checkbox. It is substrate content in each participating Self's home governance. Specifically, it is a research-use propagation permission — a specific form of propagation permission (D2.31 Dimension A) governing what may be done with a Self's contributed content once it reaches the shared substrate. A standard propagation permission specifies which content may cross the perimeter and under what conditions. A research-use propagation permission additionally specifies that the contributing Self's governance patterns may be used for the research purposes stated in the FAI configuration.

This substrate-content requirement has three architectural consequences that a verbal or procedural agreement would not produce.

Governed. The authorization is subject to the home Self's authority structure. The governing humans of the Self author the research-use propagation permission as substrate content. It is their exercised authority — their authored governance — not a transaction that bypasses governance. Any subsequent dispute about whether authorization was granted has a substrate artifact to resolve it.

Attributed. Substrate content carries provenance. The research-use propagation permission is authored content, linked to the governing humans who authored it, with the full audit trail the home governance provides. Attribution is architectural rather than procedural.

Auditable. Because the authorization is substrate content, it is inspectable at any time by any human with appropriate access. An observer reviewing whether an observational study FAI event was properly authorized can locate each participating Self's research-use propagation permission in that Self's home substrate — not in a separate consent management system, but in the same governance substrate the rest of the event's governance lives in.

The negative case is equally important: without an explicit research-use propagation permission present as substrate content in a participating Self's home governance, that Self's contributed governance patterns cannot be used for research purposes under the architecture's authorization model. Operational participation in the FAI event — contributing aspects,

receiving merged content, generating governance records — does not entail research-use authorization. The two authorizations are distinct substrate content items serving distinct purposes.

4. Research record as additional governance artifact

Standard FAI governance records — the seventeen categories established in D2.18 — document what happened in the event. They are the accountability artifacts: who participated, what was contributed, how conflicts were handled, what the event produced, what was retained and where. These seventeen categories are complete for operational accountability purposes.

An observational study FAI event produces a further artifact: the research record. The research record is not an eighteenth category within the D2.18 governance record structure. It is a separate governance artifact serving a distinct purpose — research findings rather than operational accountability. The difference is categorical: D2.18's governance record answers what happened; the research record answers what the research found.

The research record contains four components.

Research question specification. Which governance patterns were studied, and what research questions the event was designed to address. This connects the record to the research-purpose statement in the FAI configuration (§2), providing traceability from purpose to findings.

Findings. What the event's governance data revealed about the studied patterns. The research record is the artifact that carries these findings as governed content — attributed, auditable, and subject to the authority of the cross-organizational governance agreement that governs the shared substrate.

Handling specification. How research findings will be used: published with attribution, shared within a consortium, held confidential, or some combination. This specification is part of the research record itself, not a separate agreement.

Consent records. A reference from each participating Self's research-use propagation permission to the research record, confirming which event's research findings each authorization covers. This links the consent architecture (§3) to the research findings artifact.

The research record lives as shared substrate governance content for the duration specified in the cross-organizational governance agreement. Its persistence policy is a configurable dimension of the observational study FAI configuration, following the same configurability properties that apply to all FAI governance artifacts (D2.12; Paper 3 §8). When the shared substrate dissolves, persistence policy determines where the research record is retained and under whose governance.

5. Anonymization governance

Many observational study FAI events will involve organizations that are willing to contribute governance pattern data for research purposes but are not willing to have their participation or governance patterns publicly attributed. This is a standard and legitimate configuration. The architecture addresses it through the cross-organizational governance agreement (D2.34).

The cross-organizational governance agreement for an observational study FAI event must specify, for each participating Self: whether contributed governance patterns may be attributed in published research findings, whether anonymization is required, and at what level of granularity attribution or anonymization applies.

Attribution-permitted configuration. The participating Self's research-use propagation permission explicitly permits attribution of contributed governance patterns in published or shared research findings. Research records in this configuration may identify the contributing organization when reporting findings. This configuration is appropriate when the organizational context of governance patterns is material to the research findings and the organization is willing to be identified.

Anonymization-required configuration. The participating Self's research-use propagation permission requires that contributed governance patterns be anonymized before research findings are published or shared. Research records in this configuration use de-identified representations of the contributing organization's governance patterns. The cross-organizational governance agreement specifies the anonymization method and the standard against which anonymization is evaluated.

Mixed configurations — where some participating Selves permit attribution and others require anonymization — are architecturally valid. The cross-organizational governance agreement carries the per-Self specification; the research record carries the corresponding anonymization status for each included data set.

The anonymization specification is governed in the same sense as all other dimensions of the FAI configuration: it is substrate content under joint authority, authored before the event begins, auditable through the shared substrate governance artifacts.

6. Connection to Claim 6 empirical validation

Paper 3's Claim 6 is rendered in calibrated-humility register throughout §9. The architectural commitment — that accumulated FAI events across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope — is defended as a design-pattern claim, not as an empirical finding. The paper is explicit: population-scale CKS-governed Self deployments do not yet exist; specific population-scale dynamics await empirical observation; empirical validation at scale is downstream work (Paper 3, §9.1, §11.10).

Observational study FAI events are the governed mechanism through which that downstream empirical validation could proceed. This is not a speculative extension of the architecture; it is a direct application of the architecture to the research context the paper itself identifies as future work. The connection has a specific shape.

Claim 6 projects that accumulated FAI events drive collective evolution dynamics at population scope. Empirically validating this projection requires data about governance patterns across many Selves participating in FAI events over time. Observational study FAI events, structured as the architecture specifies, produce exactly that data — under the architecture's full accountability properties. The research data is governed (produced in governed events), attributed (each contributing Self's governance patterns carry home-substrate provenance), and auditable (research records live as shared substrate governance artifacts with full traceability back to each event's configuration).

Population-scale accumulation of observational study FAI events would produce a research corpus about governance evolution dynamics that is itself architecturally grounded: the research data has the same provenance, accountability, and auditability properties as the governance patterns it describes. This is a property observational research conducted through ungoverned means — surveying organizations about their governance practices, for instance — cannot provide.

This note records the observational study FAI configuration as a formalized derivation in the calibrated-humility register appropriate to Claim 6's scope. The architectural path to empirical validation is specified; whether that path will be traveled, at what pace, and what it will find are questions downstream research will answer.

7. Anti-pattern: ungoverned research use

The anti-pattern for observational study FAI governance is ungoverned research use: using FAI governance records — or any substrate content produced in FAI events — for research purposes without explicit research-use authorization from each contributing Self's governance.

Ungoverned research use takes several forms.

Form 1 — Retroactive repurposing. FAI governance records are produced in an operational coordination event, and a researcher subsequently repurposes them as research data without obtaining research-use authorization from the participating Selves. The governance records were produced for operational accountability, not for research. The Selves' governing humans authorized an operational event; they did not authorize research use of its artifacts. Using those artifacts for research without subsequent explicit authorization violates the authorization model.

Form 2 — Informal consent substitution. A researcher obtains verbal or informal agreement from a representative of a participating organization, and proceeds to use that organization's

contributed governance patterns for research without authoring a research-use propagation permission as substrate content in that Self's home governance. The informal agreement does not satisfy the architecture's authorization model: it is not governed, not attributed as substrate content, and not auditable through the home substrate.

Form 3 — Aggregate anonymization as authorization. A researcher argues that because individual contributing Selves' patterns will be anonymized in published findings, no per-Self authorization is needed. Anonymization affects what appears in published outputs; it does not substitute for authorization over the data itself. The research-use propagation permission governs the use of a Self's contributed patterns in the research process — including collection, analysis, and storage — not only what appears in published outputs.

In each form, the underlying violation is the same: substrate content that belongs to the participating Selves is used for a purpose the governing humans did not authorize as substrate content. The three properties that make research-use authorization meaningful — governed, attributed, auditable — are exactly the properties that ungoverned research use bypasses.

8. Operational test

An observational study FAI event satisfies the architecture's research governance requirements if and only if an observer with appropriate access can find all of the following.

Test element 1 — Research-use authorization in each participant's home substrate. For every participating Self, the observer can locate, in that Self's home governance substrate, a research-use propagation permission explicitly authorizing the research purposes stated in the FAI configuration. The permission is substrate content with home-substrate provenance. No participating Self's governance patterns are used for research absent this substrate content.

Test element 2 — Research record in the shared substrate governance artifacts. The observer can locate, in the shared substrate's governance artifacts, a research record that is distinct from the seventeen D2.18 governance record categories. The research record specifies which governance patterns were studied, what research questions the event addressed, how findings will be handled, and references each participating Self's research-use propagation permission.

Test element 3 — Anonymization specifications in the cross-organizational agreement. For each participating Self, the observer can locate, in the cross-organizational governance agreement, the per-Self anonymization specification — whether attribution is permitted or anonymization is required, and at what granularity. The specification is present for every Self, not only for those that require anonymization.

An observational study FAI event that passes all three test elements satisfies the research governance requirements. An event that fails any element has an authorization or

documentation gap that the operational test surfaces for resolution.

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